The Local Economic Impact of Data Centers

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Abstract

A hyperscale data center can cost more than a billion dollars while employing only a few dozen people. Using a registry of 341 hyperscale facilities, we ask whether this investment generates local agglomeration spillovers and find little evidence that it does. Nighttime lights rise sharply at the parcel when construction begins, and the increase fades within five kilometers. Announced projects of similar scale that were never built show no comparable change. Advertised salaries and new-firm registrations do not rise, and we can rule out increases above 5.5 and 1.6 percent. Hiring and supplier estimates are also near zero but less precise. County data show no credible growth in related industries. Rent estimates are positive but imprecise. Compute firms locate near data centers, but most were already there before the facility opened.

Keywords: data centers, agglomeration, economic geography, place-based investment.

JEL Codes: R11, R23, O33, H71.

1 Introduction

The physical infrastructure behind digital services has become a major component of US capital investment. Data centers consumed roughly four percent of US electricity in 2023, and the rise of artificial intelligence has accelerated new construction [Shehabi et al., 2024, International Energy Agency, 2025]. A hyperscale campus can cost more than a billion dollars and draw hundreds of megawatts of continuous power, yet it may employ only a few dozen people on site [Alvarez et al., 2026]. At least thirty-five states now offer tax incentives specifically for data centers, with cumulative subsidies approaching twenty billion dollars.[1] These subsidies rest partly on the same promise of local economic gains that motivates place-based incentives more broadly [Slattery and Zidar, 2020, Kline and Moretti, 2014]. We study whether data-center entry produces those gains.

The natural benchmark comes from the literature on large plant openings. Greenstone et al. [2010] compare counties that won a new plant with runner-up counties that lost it and find that incumbent establishments become more productive in the winning counties. They attribute the gain to a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. From a policy perspective, this productivity gain helps motivate place-based investments. At the same time, later work shows that winner-versus-loser estimates are sensitive to the choice of comparison counties [Patrick, 2016], while the average large plant produces little evidence of productivity spillovers [Patrick and Partridge, 2022]. Plant openings therefore do not provide a settled benchmark, even though manufacturing offers the strongest traditional case for agglomeration.

Data centers separate the location of capital from many of the workers, suppliers, and customers that use it. This contrast lets us ask whether a large investment generates local spillovers. Labor pooling, input sharing, and knowledge spillovers can all make proximity productive [Duranton and Puga, 2004, Ellison et al., 2010], and these channels allow a large employer to generate a local multiplier [Moretti, 2010]. A data center may engage each channel only weakly. It employs dozens of people rather than thousands, purchases many inputs from regional or national suppliers, and delivers its output over the network to distant customers. The scale of the investment may therefore reveal little about the size of its local spillovers.

Data centers do not locate at random. Operators screen sites on power, land, and fiber, so places that host a facility differ from places that do not in ways that also shape local outcomes. A credible instrument for this siting decision is difficult to find. We therefore organize the analysis around two complementary facility-level designs. The within-site design compares changes near a completed facility with changes farther from the same site. At completed sites, we date construction from satellite-detected land clearing and restrict the capital analysis to facilities without an earlier data center within two kilometers. This comparison absorbs characteristics shared by the surrounding area, and we use it only when the outcome has a flat pre-trend. The built-versus-near-miss design compares completed facilities with 84 announced projects of hyperscale size that were never built, 52 of which have satellite coverage. These projects crossed an observable announcement and siting

[1] Authors' tally from the Good Jobs First Subsidy Tracker [Good Jobs First, 2024], which records state and local data-center subsidy awards and is one of the sources for our facility registry (Section 2).

threshold, although they tend to be newer, farther from existing data-center clusters, and less likely to have a named hyperscale sponsor. We therefore do not treat them as otherwise identical counterfactuals. County panels provide descriptive evidence on industry outcomes, although pre-period trends limit what those panels can establish.

Construction visibly transforms the site. At isolated sites, nighttime lights rise by 38 percent within the first kilometer when construction begins, an increase that fades to near zero by five kilometers [Henderson et al., 2012]. Daytime imagery confirms that vegetation clears at the same time, while announced projects that were not built show no increase in lights. The remaining question is whether this activity spreads beyond the parcel.

We find little evidence of a broader response. Advertised salaries and new-firm registrations do not rise, with confidence intervals that exclude increases above 5.5 and 1.6 percent, respectively. Business applications also do not rise. Hiring and supplier estimates are less precise, and the job-posting interval permits an increase of up to 17 percent. Workplace employment begins rising before the facility opens rather than breaking at the event. County data show a 27 percent increase in data-processing establishments, which include the facility itself, while broader information and professional services were already growing beforehand.

An industrial cluster could develop around data centers if compute-intensive firms move toward new facilities over time. And, in fact, compute-using firms are indeed located closer to data centers than other firms, but this is mainly because both are concentrated in metropolitan areas. Moreover, 69 percent of nearby compute firms were founded before the nearest data center opened. Together, the counts and the timing suggest that their proximity reflects shared metropolitan advantages rather than a compute cluster created by the data center.

Other local outcomes may adjust even without a broad worker or firm response. Nearby rents may rise by a few percent, but the estimates are imprecise. We therefore treat the rent evidence as suggestive. Neither multifamily permitting nor net migration rises. School property-tax revenue and county highway spending have positive point estimates, but neither is statistically significant after correcting for the six fiscal outcomes we test. We find no robust change in foot traffic or commercial spending, and residential electricity prices do not rise. The evidence leaves open modest effects on rents or public revenue, but not broad adjustment in population, construction, consumer activity, or electricity prices.

Taken together, these results distinguish a large local investment from a broad local response. Existing county studies find positive effects on employment, establishments, income, construction, and house prices [Alvarez et al., 2026, Yue and Zeng, 2026]. Our finer data separate activity at the facility from changes among nearby workers, suppliers, and firms and establish whether compute users arrived before or after the data center. At this scale, the sites are transformed, but we find little evidence of a new local cluster. This matters as US place-based investment shifts from labor-intensive plants to labor-light digital capital.

2 Data

Hyperscale data-center registry. We construct a master registry of US data centers from several sources. We begin with the Open Source Data Center Atlas, produced by Pacific Northwest National Laboratory's IM3 project, which identifies 1,242 facilities from OpenStreetMap footprints tagged as data centers and provides coordinates along with square footage and operator when available. We supplement the atlas with operating dates from operator press releases, state economic-development announcements, the Good Jobs First subsidy tracker, and trade publications. Targeted searches of state records, specialty wholesale operators, artificial intelligence operators, and real-estate investment trust filings add facilities that are missing from the atlas. The resulting master registry contains 1,494 facility records. Our analysis begins with the 341 records that we classify as hyperscale based on operator identity. This category includes Amazon, Microsoft, Google, Meta, Apple, Oracle, and a hand-curated group of large specialty and artificial intelligence campuses. We do not impose a megawatt threshold because capacity is not consistently observed. Of the 341 hyperscale records, 334 have an analysis date within the 2012–2024 VIIRS window and 315 have a high- or medium-confidence date.

For the built-versus-near-miss comparison, we separately assemble 84 announced projects that were not built. Of these, 52 have the coordinates and satellite coverage needed for the nighttime-lights design, and 49 are sufficiently far from operating campuses to enter the isolated capital sample. Section 3 describes the sponsors of these projects and shows that the construction-year estimate is 72 percent when we restrict the control group to projects with a named hyperscale tenant, compared with 70 percent in the full isolated sample.

Table 1: Hyperscale sample construction

Sample restriction	N
Panel A: Hyperscale registry	
Hyperscale facility records	341
with an analysis date in 2012–2024	334
with high- or medium-confidence date	315
with satellite-detected construction date	87
Panel B: Capital-footprint design	
Announced hyperscale projects that were not built	84
with coordinates and satellite coverage	52
Isolated, construction-dated built sites	**43**
Isolated announced near-miss sites	**49**

Notes: The 341 hyperscale records are drawn from the broader 1,494-record registry and include the six major cloud operators and hand-classified large specialty and artificial-intelligence campuses. High- and medium-confidence dates exclude projected openings, unverified planned-completion dates, and unresolved source conflicts. The capital design also requires a satellite-detected construction date and no earlier data center within two kilometers.

Nighttime lights. We extract monthly mean radiance from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band Monthly Composite from April 2012 through December 2025. Around each facility, we measure radiance in concentric rings of 0–1, 1–2, 2–5, 5–10, 10–25, and 25–50 kilometers and aggregate the monthly observations to annual means. We define a facility as isolated when no data center within two kilometers opened earlier, and the headline capital-footprint estimate uses this isolated, construction-dated sample.

Local labor, housing, and spending panels. Several proprietary data products measure local activity around each site. WageScape provides job postings and advertised salaries by geocoded location [WageScape, 2022]. LinkUp Job Records provide employer-level postings used in the firm analysis [LinkUp, 2025]. RentHub Rental Data provide monthly median asking rents by ring [RentHub, 2022]. SafeGraph Spend Patterns measure consumer spending [SafeGraph, 2022], and Advan Weekly Patterns measure foot traffic [Advan Research, 2025]. We analyze rents, spending, and foot traffic in a within-site ring design dated to the operating year, with state-by-month fixed effects. The posting and salary panels extend to 150 kilometers, providing the comparison area used in the labor analysis (Section 3.2).

Coverage differs across these sources, so each analysis uses the largest subset of the 315 cleanly dated hyperscale records for which its outcome and design are available. As a robustness check, Appendix Table D.2 restricts the national facility-ring outcomes to a common universe of 125 hyperscale sites. The construction, two-state business-registration, and county analyses retain their design-specific samples because they require different dates or use different geographic units.

County employment. The county-level analysis uses employment and establishment counts from the Quarterly Census of Employment and Wages. We retain the Bureau of Labor Statistics disclosure flag because the public files code suppressed employment as zero even though the underlying value is unobserved. The analysis covers data processing (NAICS 518), telecommunications (517), web and information services (519), IT consulting (5415), and professional, scientific, and technical services (541). We end the 519 series in 2021 because the 2022 NAICS revision moved major components of that sector to 513 and 516. Section 5.5 reports the disclosure rates and pre-trend diagnostics that limit causal interpretation of the county results.

Instrument-search inputs. The instrument search draws on predetermined county features, namely 345 kV transmission-substation density, the InterTubes long-haul fiber network [Durairajan et al., 2015], the 1980 county share of urban college population (NHGIS), station-level climate normals (CustomWeather), and the EIA-860 generation fleet (Appendix B).

2.1 Dating construction and operations

We preserve announcement, groundbreaking, construction, planned-completion, and operating dates as distinct events. Operating dates reported by operators or public agencies receive high confidence. Less detailed sources receive medium confidence when the year agrees with an independent registry or satellite evidence. We classify announcement-based projections, planned completion dates, and unresolved conflicts as low confidence and exclude them from analyses that require a reliable date. This process produces 315 high- or medium-confidence hyperscale dates, 19 low-confidence dates, and 7 facilities dated outside the 2012–2024 window.

For a smaller sample, we recover the start of construction from daytime satellite imagery. A hyperscale build begins with land clearing, which appears in Landsat and Sentinel-2 imagery as a sustained decline in the parcel's vegetation index, a rise in its built-up-surface index, or both. Vegetation remains stable before the detected construction year and then falls sharply. We validate this timing with Federal Aviation Administration obstruction-evaluation filings, which are recorded independently of the satellite data. The imagery sample contains 87 hyperscale facilities. The FAA records match 64 facilities within 500 meters and 119 within one kilometer (Section 3.5). Together, these independent dates support the satellite construction timing used in the capital design.

3 Research Design

Data centers do not locate at random. Operators screen locations on electricity prices, available grid capacity, large parcels of low-cost land, fiber proximity, and tax policy. Counties that attract a data center therefore differ from other counties in ways that may also affect the local economy, making a simple cross-county comparison difficult to interpret. Appendix B documents this selection by testing a range of predetermined siting predictors as potential instruments. Power infrastructure, industrial land, and fiber strongly predict siting, but each is also correlated with local industrial conditions. Climate and historical human capital pose fewer obvious exclusion concerns, yet neither predicts hyperscale locations.

We organize the analysis around two complementary facility-level designs. The within-site design compares changes close to a completed facility with changes farther from the same site, thereby absorbing time-invariant siting differences. The built-versus-near-miss design asks whether changes around completed facilities exceed those around announced projects that were not built. We report both where the data allow. Unlike the runner-up counties in Greenstone et al. [2010], the announced projects are not known finalists in the same site-selection process, so we do not assume that they are balanced with built sites. County panels separately describe aggregate industry changes, although suppression and pre-opening trends prevent us from interpreting most of those estimates as causal effects on related employment.

3.1 Announced near-miss facilities

An announced project that reached the siting stage but was never built provides an external comparison for a completed facility. Both groups crossed an observable announcement and siting threshold, although projects that proceed may still differ from those that do not. We identify 84 announced projects that were later cancelled and were large enough to qualify as hyperscale. The sample comes from trade publications, regulatory filings, and operator announcements. Of these, 52 projects have the coverage required for the nighttime-lights design. For each outcome, we estimate

$$y_{i,r,t} = \sum_k \left[\beta_k \, \mathbf{1}[t - t_i^c = k] \times \text{Near}_{i,r} \times \text{Built}_i + \delta_k \, \mathbf{1}[t - t_i^c = k] \times \text{Near}_{i,r}\right] + \alpha_{i,r} + \lambda_t + u_{i,r,t}, \quad (1)$$

where Built_i identifies completed facilities, t_i^c is the satellite-detected construction start for built sites and the announcement date for cancelled sites, $\alpha_{i,r}$ is a facility-by-ring fixed effect, and λ_t is a calendar-year fixed effect. In the capital analysis, $y_{i,r,t}$ is the log of mean radiance, and the path of β_k measures the change around built facilities relative to the change around cancelled sites. Timing is not symmetric across the two groups because cancelled projects lack an observed or consistently reported construction start. Announcements usually precede construction by at least a year, placing some announcement-related activity for built sites in their own pre-period and, if anything, attenuating the estimated break. This limitation is less consequential for the capital outcome because a parcel that is never built does not light up, regardless of why the project was cancelled.

The announced near misses also differ in sponsor commitment. Fourteen of the 84 projects had a named hyperscale operator at cancellation, 19 were sponsored by established data-center operators without a named tenant, and the remaining 51, roughly 60 percent, were speculative developments led by real-estate sponsors. Projects without a signed tenant or confirmed grid allocation are less informative about the counterfactual for a committed facility. Even so, restricting the capital comparison to the nine isolated projects with a named hyperscale operator produces a 72 percent increase in the groundbreaking year, compared with 70 percent in the full isolated sample. The named-sponsor subset has a somewhat less stable pre-period and is too small to support longer-run estimates. More generally, many cancellations followed local opposition, so the near-miss sites may be located in less development-friendly places.[2] These limitations prevent us from treating the near misses as balanced runner-up sites. We therefore use them alongside the within-site comparison below. Built sites are dated from satellite-detected land clearing, whereas near-miss timing comes from less reliable schedules in filings and trade publications. Independent regulatory filings validate the built-site dates (Section 3.5).

Table 2 describes these differences in the exact capital-footprint sample. Built and near-miss sites have similar local populations and access to high-voltage infrastructure, but they differ substantially in event year, baseline radiance, proximity to an earlier data center, region, and sponsor commitment. Facility-by-ring fixed effects absorb the level differences, while the pre-event path reveals whether the inner and outer rings were already diverging. The imbalance in timing remains a concern. The near-miss group contains only one to five projects per calendar year before 2020

[2]Because most near-miss projects were announced late in the sample, the near-miss group contributes few post-event observations. Coverage at event years zero through four is 48, 21, 5, 3, and 1 sites, so the built-versus-near-miss path at longer horizons is identified primarily from changes among built sites relative to the common pre-event trend rather than from post-period movement among near-miss sites. Section 4 quotes magnitudes only through the year after groundbreaking for this reason.

and 21 to 49 thereafter, whereas 26 of the 43 built sites break ground in 2017 or 2018. Section 4 therefore reports results by construction cohort and shows that the within-site design, which does not use near-miss controls, reproduces the same cohort pattern. The observable imbalances prevent us from interpreting the near misses as conventionally balanced runner-up sites.

Table 2: Built and announced near-miss sites in the capital-footprint sample

Characteristic	Built mean	Near miss mean	Standardized difference
Event year	2018.81	2024.37	-4.21
Log population within 10 km	10.54	10.42	+0.06
Pre-event log radiance, 0–1 km	0.54	2.26	-1.28
Log distance to earlier data center	2.35	3.64	-1.19
Log distance to ≥230 kV substation	1.87	1.90	-0.04
Log distance to long-haul fiber	2.15	1.87	+0.29
Share Northeast	0.00	0.06	-0.36
Share Midwest	0.23	0.41	-0.38
Share South	0.35	0.43	-0.16
Share West	0.42	0.10	+0.76
Share with named hyperscale sponsor	1.00	0.18	+2.95

Notes: Exact isolated-site sample used at event year zero in Figure 1, namely 43 built sites and 49 announced near-miss sites. Built sites are dated by satellite-detected construction start. Near-miss sites are dated by announcement. The standardized difference is the difference in group means divided by the square root of the average of the two group variances, so a value of one means the groups differ by one pooled standard deviation. Distances are in kilometers before applying $\log(1+x)$. Population is based on 2020 Census tract population centroids. High-voltage access uses substations rated at least 230 kV. Long-haul fiber uses the InterTubes conduit map. The table is a diagnostic, not a claim that the groups are balanced.

3.2 Within-site rings

The near-miss comparison is available when the same outcome can be reconstructed around never-built sites. For outcomes observed only around operating facilities, the within-site design compares an inner ring around a facility with the 25 to 50 kilometer ring around the same site,

$$\log(1+\mathrm{NTL}_{i,r,t}) = \sum_k \beta_k \mathbf{1}[t - t_i^c = k] \times \mathrm{Near}_{i,r} + \alpha_{i,r} + \gamma_t + \varepsilon_{i,r,t}, \quad (2)$$

where i indexes the facility, r the ring, t the year, and t_i^c the facility's construction-start year. The indicator $\mathrm{Near}_{i,r}$ marks the inner ring, $\alpha_{i,r}$ is a facility-by-ring fixed effect, and γ_t is a calendar-year fixed effect. The facility-by-ring fixed effect absorbs any time-invariant difference between a site's inner and outer ring, including siting preferences, local industrial structure, and access to grid and fiber infrastructure. The coefficients are therefore identified by changes in the inner ring relative to changes in the outer ring of the same site.[3]

Two changes address the pre-trend present in a simpler version of the design. First, we date the event to construction rather than operations, so that activity during the building phase does not appear in the pre-period (Section 2.1). With this timing, inner-ring brightness is flat before groundbreaking and rises when construction begins. Second, the headline sample includes only isolated facilities with no earlier data center within two kilometers. A new facility built beside an existing campus starts with a brighter inner ring and can create a spurious pre-trend. In the isolated sample, the event study is flat before construction, rises by 38 percent in the construction year, and continues to build over the following years. A separate linear trend fitted to each site's inner-minus-outer path during the four pre-construction years produces the same 38 percent observed-minus-predicted change, with a campus-bootstrap 95 percent interval of 15 to 71 percent.

We interpret ring estimates only when the outcome is stable before construction. The capital footprint, rents, advertised wages, and firm entry satisfy this condition. Workplace employment does not, and we place no interpretive weight on its inner-versus-outer estimate. Labor and supplier effects may also extend beyond the immediate site. For advertised wages, job postings, and supplier postings, we therefore compare the pooled 0 to 25 kilometer area with locations 50 to 150 kilometers away. This geography allows for effects across a commuting zone while keeping the comparison area outside the immediate data-center cluster.

3.3 Demand-side outcomes

We apply the same within-site comparison in equation (2) to residential rents, consumer spending, and foot traffic, dating these outcomes to the operating year rather than the start of construction. Because the monthly panels span 2020 to 2023, we replace the calendar-year fixed effect with state-by-month fixed effects that absorb differences across states in work-from-home patterns and the urban recovery. We also report built-versus-near-miss estimates where the data permit. Rents rise modestly at built sites relative to near misses, while foot traffic shows no robust change and commercial spending is unchanged in either design. Section 6 uses these estimates to examine who bears the local incidence of entry.

3.4 Employment and county-industry diagnostics

Local labor demand is harder to identify than the capital footprint, so we measure it at two geographic scales. The local analysis counts vacancies advertised near each facility and uses the 50 to 150 kilometer comparison area described above. A second analysis moves to the county level, where the data center's own workers and any related firms need not fall inside a single ring. Using

[3] With two rings, replacing γ_t with a ring-by-year term would absorb the average inner-versus-outer gap in each calendar year and leave the event-study coefficients identified only from differences in construction timing across sites. We use the calendar-year specification throughout. Section 4 reports the estimate under the log of radiance as well, so that its magnitude can be compared with the near-miss design.

employment and establishment counts by industry from the Quarterly Census of Employment and Wages, we estimate

$$\log(1 + \text{Emp}^s_{c,t}) = \sum_k \theta^s_k \, \mathbf{1}[t - t^o_c = k] \times \text{DC}_c + \mu_c + \lambda_t + u_{c,t}, \tag{3}$$

where s indexes industry, c county, and t^o_c the county's first data-center year. The specification includes county and year fixed effects. We also estimate cohort-by-event-time effects using not-yet-treated counties as controls and state-cluster bootstrap inference, following Callaway and Sant'Anna [2021]. Two data features constrain the interpretation. Depending on the sector, 27 to 68 percent of detailed employment cells are suppressed, and establishment counts in several industries begin rising before opening. The county panels therefore cannot distinguish growth caused by the facility from a continuation of those earlier trends, nor can they establish a precise zero. Section 5.5 reports these diagnostics and results in full.

3.5 Credibility of the capital-footprint design

The capital-footprint estimate is stable across the standard checks for a ring design. Dropping each facility in turn leaves the coefficient in a narrow range, and excluding the campus itself and using the 1 to 2 kilometer ring as the inner comparison produces the same spatial decay. Because hyperscale facilities often share a campus, we cluster standard errors at the campus rather than the facility level. This widens the confidence interval but leaves the effect statistically significant.

Federal Aviation Administration obstruction-evaluation filings provide an independent check on construction timing. We match facilities by location to these filings, whose timestamps are recorded before any radiance response and do not use satellite information. Re-dating sites with these filings produces a capital footprint that is positive and grows after filing. Because an FAA filing usually precedes physical construction, the comparison establishes the ordering of events rather than an exact alternative groundbreaking date.

Both the ring and near-miss designs use staggered construction dates. As an additional check, we compare each construction cohort only with never-built sites observed in the same calendar years (Section 4). This cohort-robust estimate is imprecise and fails its lead test, so we treat it as an inconclusive diagnostic rather than corroborating evidence.

The rent analysis requires an outcome-specific adjustment. Between 2020 and 2023, rents rose faster near metropolitan cores than in surrounding areas. Because a data center's inner and outer rings occupy different points on this gradient, the ring comparison may capture the metropolitan gradient rather than the facility. Section 6 therefore adds a synthetic control for each facility from population-matched locations without a data center. This comparison absorbs common growth and isolates the additional change at the data-center location.

Table 3 summarizes which design carries the interpretation for each outcome. No single comparison is credible for every outcome.

Table 3: Outcome estimates across research designs

Outcome (hyperscale)	Built vs. near miss	Spatial comparison	County evidence
Capital (nighttime lights)	**+70% ($p < 0.01$)**	+38% ($p < 0.01$)[a]	–[b]
Residential rent	+4.3% ($p = 0.23$)	**+2 to +5%**[c]	–[b]
Advertised salary	−11.6% ($p = 0.54$)	**+1.3%** ($p = 0.53$)[d]	–[b]
Job postings	+4.0% ($p = 0.75$)	**+5.2%** ($p = 0.35$)[d]	–[b]
Supplier postings	–[e]	**+5.9%** ($p = 0.61$)[d]	–[b]
Foot traffic	**−12.0%** ($p = 0.12$)[f]	excluded[g]	–[b]
Consumer spending	**+4.6%** ($p = 0.60$)	+8.5% ($p = 0.50$)[h]	–[b]
Firm entry (SoS registrations)	–[e]	**−0.5%** ($p = 0.62$)[i]	+1.4% ($p = 0.49$)[j]
Workplace employment	–[e]	excluded[g]	inconclusive[k]

Notes: Post-event estimates for hyperscale facilities at the innermost ring each design supports. Bold marks the single design used to interpret each outcome. The built-versus-near-miss capital estimate is the isolated-sample construction-year jump. The two groups are not balanced on several observable characteristics (Table 2), and magnitudes at longer horizons rest on few near-miss controls (Section 4).

[a] Agrees with the near-miss estimate once both use the same transform. The gap is $\log(1 + x)$ against $\log(x)$, not the comparison group (Table 4).

[b] Not estimated at the county level. There is no county analog of the ring outcome.

[c] Synthetic-control headline. The within-site estimates (+6.6 to +9.4%) are an upper bound (Section 6).

[d] Pooled 0–25 km area against a 50–150 km comparison area, allowing labor-market effects to extend beyond the immediate site (Section 3.2).

[e] No panel around near-miss sites exists for this outcome (business registrations and LODES cannot be reconstructed around never-built sites at ring scale).

[f] Attenuates to −3.0% ($p = 0.68$) on the balanced panel, so we do not interpret it as reliable evidence of displacement.

[g] Excluded because the pre-opening coefficients show an inner-versus-outer trend that the ring design cannot separate from an effect (Section 3).

[h] Reported with caveats in Section 6 and passes a random-donor comparison.

[i] Hyperscale-only sample using updated opening dates. The pooled all-archetype estimate is +0.7% ($p = 0.24$).

[j] Census Business Formation Statistics, national county design.

[k] Data-processing establishments rise with direct entry. Detailed QCEW employment is heavily suppressed, while establishment counts in several downstream sectors rise before opening. Estimates that preserve the suppression flags and use not-yet-treated controls therefore do not identify either a downstream gain or a precise zero (Section 5.5).

4 The capital footprint

4.1 Within-site ring design

The within-site estimate compares the inner ring around a facility with the 25 to 50 kilometer ring around the same site, using the construction start recovered from satellite-detected land clearing (Section 2.1). Two aspects of this design determine how we estimate the effect. First, a hyperscale campus takes two to three years to build. Dating the event to the operating year would place the construction phase, when the parcel is already changing, in the pre-period and misclassify that activity as a pre-trend. Observing construction and operations separately avoids this problem.

Second, the headline sample excludes sites built next to an existing data center. Roughly half of hyperscale facilities with a satellite construction date lie within two kilometers of an earlier facility and therefore inherit some of its brightness before construction begins. At these non-isolated sites, inner-ring brightness rises by about 50 percent during the pre-period. By contrast, the inner ring at an isolated site contains only the future parcel. Its pre-construction excess is 7 percent, and the event study is flat before groundbreaking. A joint test does not reject equality of the pre-construction coefficients to zero ($p = 0.37$), after which brightness rises by 38 percent in the construction year. Fitting a separate linear trend to each site's inner-minus-outer path over event years -4 through -1 and extrapolating to event year zero produces the same 38 percent increase, with a campus-bootstrap 95 percent interval of 15 to 71 percent.

4.2 Built versus announced near misses

Figure 1 reports the built-versus-near-miss comparison from equation (1). The comparison follows radiance within one kilometer of 43 isolated built facilities and 49 announced projects located away from operating campuses that were not built. As Table 2 shows, the two groups differ on several observable characteristics.

Before construction, the coefficients range from 8 to 11 percent below the omitted year, but they are jointly indistinguishable from zero when clustered by campus ($p = 0.24$), and their linear slope is also insignificant (+3.5 log points per year, $p = 0.22$). Radiance then rises by 70 percent in the construction year and reaches 231 percent above the near-miss sites one year later. Campus-level clustering widens the confidence intervals by roughly 60 percent, yet every post-groundbreaking coefficient remains statistically significant. The nighttime-light footprint captures activity at the parcel, including security and parking lights, substations, cooling equipment, and eventually continuous operation.

Two features determine how this benchmark should be interpreted. First, the near-miss cohort is concentrated in recent announcement years. Although 48 control sites identify the construction-year contrast and 21 identify the year-one contrast, five or fewer remain at later horizons. We therefore end the figure after year one and place no weight on subsequent coefficients.

Second, the estimate describes a campus rather than a single building. Thirty-two of the 43 isolated built sites add another building within two kilometers during the event window, which helps explain why the full-sample path continues rising through year four. The 11 sites that remain single-building projects are also much smaller, and their construction-year estimate is 7.7 percent with a standard error of 6.9 percent. Phased construction cannot explain this initial gap because no sibling building has yet been added in the construction year. Instead, the two groups differ in project scale from the outset. We therefore interpret the estimate as the physical response to a large multi-building campus, which is also the project that a jurisdiction recruits.

4.3 Reconciling the two designs

The ring and near-miss designs produce the same timing and shape, and their magnitudes align once the outcomes are placed on a common scale (Table 4). The apparent difference between the 70 and 38 percent estimates comes from the transformation of radiance rather than the control group.

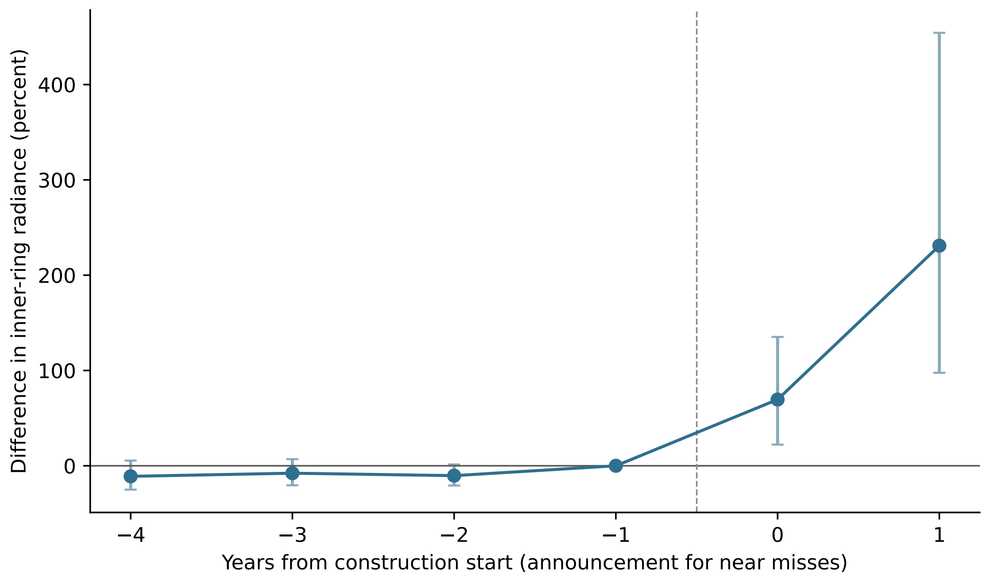

Figure 1: Capital footprint at built versus announced near-miss sites

Alt text: The event-study line is flat for four years before construction, rises sharply in the construction year, and rises further one year later. Confidence intervals widen after construction.

Notes: Event study based on equation (1). The inner-ring (0 to 1 km) log nighttime-lights path at 43 isolated built hyperscale facilities is measured relative to 49 isolated announced near-miss facilities, omitting the year before. Built facilities are dated by satellite-detected land clearing. Near-miss facilities are dated by announcement year. Standard errors are clustered on two-kilometer campus groups. Near-miss coverage is 48 sites in the construction year and 21 one year later. Table 2 reports observable differences between the groups, and Table 4 reports the estimate across designs, transforms, and construction cohorts.

The near-miss design uses the log of mean radiance, whereas the ring design uses $\log(1 + \text{NTL})$, which compresses proportional changes at parcels with very low pre-construction radiance. When the ring design also uses log radiance, it produces increases of 63 percent in the construction year and 226 percent one year later, compared with 70 and 231 percent in the near-miss design. A Poisson model of radiance levels yields 59 percent. Thus, three specifications that differ in control group and functional form place the initial footprint in a similar range.

Both designs also reveal the same gradient across construction cohorts. Among sites that broke ground by 2018, radiance rises by 97 percent in the near-miss design and 95 percent in the ring design. The corresponding estimates for sites that began in 2019 or later are 33 and 35 percent. Because the ring design does not use near-miss controls, this gradient cannot be explained by the concentration of near misses in recent years. Instead, it reflects the initial location. Earlier projects were built on much darker parcels, with median pre-event radiance of 1.1 compared with 4.5 for later projects, so the same amount of new lighting creates a larger proportional change. A cohort-robust estimator limited to 17 treated sites with never-built controls in the same calendar years yields 27 percent, with a 95 percent interval of $[-2, +92]$ percent. Because its pre-construction leads reject jointly ($p = 0.008$), we treat that estimate as inconclusive and rely on the two better-supported designs for the cohort pattern.

Restricting the near-miss group to the nine isolated projects with a named hyperscale operator yields a construction-year estimate of 72 percent, compared with 70 percent in the full isolated sample. The smaller subset has a somewhat less stable pre-period ($p = 0.15$) and cannot support longer-run estimates.

Table 4: The capital footprint across estimators and cohorts

Estimator	Construction year	Year 1	Built sites	Lead p
Panel A: Built versus announced near miss, log(NTL)				
All isolated sites	+70%	+231%	43	0.24
broke ground 2018 or earlier	+97%	+305%	26	0.41
broke ground 2019 or later	+33%	+140%	17	0.69
controls with a named hyperscale sponsor	+72%	+259%	43	0.15
Poisson on radiance levels	+59%	+153%	43	–
cohort-robust, never-treated controls	+27%	+154%	17	0.01
Panel B: Within-site rings, log(NTL)				
All isolated sites	+63%	+226%	43	0.51
broke ground 2018 or earlier	+95%	+294%	26	0.16
broke ground 2019 or later	+35%	+142%	17	0.89
Panel C: Within-site rings, $\log(1 + \text{NTL})$*, as reported in the text*				
All isolated sites	+38%	+132%	43	0.37
site-specific linear pre-trends	+38%	–	43	–
broke ground 2019 or later	+25%	+99%	17	0.90

Notes: Change in inner-ring (0–1 km) radiance at isolated hyperscale sites, in the construction year and one year later, transformed from log points to percent. Panels A and B use the same outcome, so their magnitudes are directly comparable. Panel C applies the $\log(1+x)$ transform used in the text, which compresses changes at the low pre-construction radiance of these parcels. The within-site rows use no near-miss control at all, and they reproduce the cohort gradient in Panel A, so that gradient reflects differences between early and late construction cohorts rather than the thin near-miss coverage before 2020. Early cohorts sit on darker land, with a median pre-event radiance of 1.1 against 4.5 for later cohorts, which enlarges a log change. The cohort-robust row compares each construction cohort with never-built sites observed in the same calendar years, using the 17 sites with such support, and inference is a campus cluster bootstrap with 999 draws. Its construction-year 95 percent interval is [-2, +92] percent, and its lead test rejects, so we treat that row as inconclusive rather than as corroboration. The site-specific-trend row fits a separate linear pre-trend to each site's event years -4 through -1 and reports the observed-minus-predicted event-zero change. Its campus-bootstrap 95 percent interval is [+15, +71] percent. Lead p jointly tests event years -4 through -2. Standard errors are clustered on two-kilometer campus groups throughout.

4.4 Spatial decay

The footprint is sharply localized. Its magnitude is largest within one kilometer, falls by more than half between one and two kilometers, and becomes small between two and five kilometers, with no detectable response beyond that distance. Omitting the central kilometer and treating the one-to-two kilometer ring as the inner zone produces the same decay. The nighttime-light increase therefore extends beyond the parcel itself but not beyond its immediate surroundings.

4.5 Colocation facilities as a comparison

Colocation facilities are generally smaller, multi-tenant projects that often occupy or expand existing buildings, although some are purpose-built. They produce no comparable change in nighttime lights. In the construction-dated within-site design for the first five kilometers, the construction-phase estimate is -1.0 percent ($p = 0.65$) across 21 colocation facilities, compared with $+8.0$ percent ($p = 0.02$) across 43 hyperscale facilities. The colocation estimate during operations is also near zero (-1.4 percent, $p = 0.74$). We therefore focus the remaining analysis on hyperscale facilities and report colocation results where the data support a meaningful comparison.

5 Test of agglomeration

Section 4 shows that hyperscale construction creates a large, spatially concentrated capital shock. We now ask whether that investment generates the local spillovers historically associated with large plant openings. Greenstone et al. [2010] organize these spillovers around a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. Each can raise productivity when firms cluster [Combes et al., 2012], and we test each channel at the facility level. Because labor-market effects may extend beyond the immediate site, the labor outcomes compare the pooled 0 to 25 kilometer area with locations 50 to 150 kilometers away (Section 3.2). Colocation facilities provide an additional comparison, although they are located in denser markets than hyperscale campuses.

Figure 2 places the preferred estimates and confidence intervals on a common scale. Advertised salaries and firm entry are estimated precisely enough to rule out modest increases. The job-posting interval is wider, while supplier postings are too sparse to provide a meaningful bound. Appendix Table D.1 reports the confidence intervals and minimum detectable effects underlying the figure.

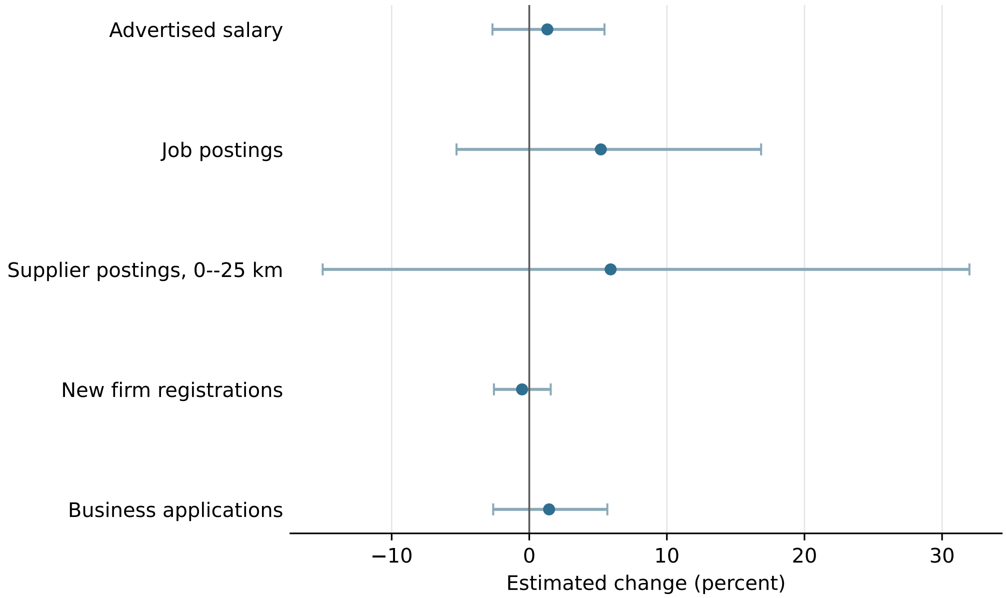

Figure 2: Precision of the agglomeration-channel estimates

Alt text: Horizontal confidence intervals for five estimates. Salary, firm-entry, and business-application intervals are narrow around zero. Job-posting and supplier intervals are wider and include sizable positive effects.

Notes: Preferred estimate for each outcome with 95% confidence intervals, transformed from log points to percent. Salary, job postings, and supplier postings compare the pooled 0 to 25 km zone with a 50 to 150 km control. Firm registrations use the Virginia and Texas within-site trend break. Business applications use the national county trend break.

5.1 Labor pooling

If a data center deepens the local labor pool, hiring and wages should rise nearby. Across 120 hyperscale facilities, job postings increase by 5.2% ($p = 0.35$), with a 95 percent interval of $[-5.3, +16.8]$ percent. The built-versus-near-miss estimate is similar at +4.0% ($p = 0.75$), with a flat pre-period ($p = 0.96$). Poisson counts, a long difference that conditions on pre-period volume, and construction dating also produce estimates near zero.

The interval is not a precise zero. Before opening, the median hyperscale facility has roughly 760 advertised vacancies per year within five kilometers. The upper bound of 17 percent corresponds to about 130 additional postings annually. Because an operating campus employs only a few dozen people, the design cannot isolate the facility's own hiring from a modest local multiplier. The confidence interval rules out a large hiring boom but allows smaller gains.

Advertised salaries provide a tighter test. Across 119 facilities, the mean log salary of nearby postings rises by 1.3% ($p = 0.53$), with a 95 percent interval of $[-2.7, +5.5]$ percent. Because the panel is concentrated in posted salaried positions, this estimate is most relevant for white-collar workers and may not capture compensation in the construction trades. The share of postings that report a salary is unchanged at opening ($p = 0.43$), making differential disclosure an unlikely explanation for the salary estimate. Workplace employment from the Longitudinal Employer-Household Dynamics data faces a different problem. Employment within five kilometers rises by 8 percent, but the increase begins before opening, and detrending reduces the opening-year change to -0.3%.

5.2 Supplier linkages

The second channel operates through local suppliers. We identify postings in the industries that build and equip data centers, including construction and specialty trades, electrical-equipment wholesale, architectural and engineering services, and machinery manufacturing. Relative to the 50 to 150 kilometer control, supplier postings within 25 kilometers rise by 5.9% ($p = 0.61$), with a 95 percent interval of $[-15, +32]$ percent. Because these postings form a thin series, the interval is too wide to distinguish a substantial increase from a decline. Supplier postings therefore do not establish whether local demand for these industries rises.

5.3 Knowledge spillovers and firm entry

The third channel is knowledge exchange among co-located firms. Appendix A maps firms that make intensive use of computing capacity. These firms are closer to data centers than the average firm, but much of the relationship reflects a shared metropolitan location, and 69 percent of nearby compute firms predate the nearest facility. Exurban data centers, by contrast, often have few compute firms nearby. Although these descriptive patterns do not identify an opening effect, they do not show a compute cluster forming around the facility.

Firm entry also does not rise. We geocode secretary-of-state business registrations in Virginia and Texas to the facility rings and find no average break at opening. Within five kilometers, registrations rise by 0.7% ($p = 0.24$) in the pooled sample and fall by 0.5% ($p = 0.62$) around hyperscale facilities. A national county design using Census Business Formation Statistics yields a similarly small estimate of $+1.4\%$ ($p = 0.49$). The upper bounds of the corresponding 95 percent intervals are 1.6 and 5.7 percent. These results rule out systematic entry effects above those bounds, while allowing that individual locations may respond differently.

5.4 Incumbent productivity

Greenstone et al. [2010] find total factor productivity gains of roughly 12 percent among incumbent establishments. We do not observe plant-level productivity, so we instead examine advertised salaries and hiring at firms that existed before the data center, two margins through which a large productivity gain might appear. The salary interval has an upper bound of 5.5 percent. It would rule out a 12 percent productivity gain only if at least 45 percent of that gain passed through to posted salaries. With lower pass-through, such a gain would remain inside the interval. The salary result therefore bounds the wage response, not productivity itself. Incumbent hiring also shows no positive break at opening, but the pooled and colocation samples have a downward pre-trend that the design cannot fully remove. We treat this hiring pattern as corroborating evidence rather than a causal estimate and do not claim to measure productivity directly.

5.5 What county-industry data can establish

County-industry data extend the analysis beyond job postings, but the detailed Quarterly Census of Employment and Wages cells create two problems in this setting. The first is disclosure. The Bureau of Labor Statistics suppresses employment when publishing a value could reveal an individual employer, and the public files record these unobserved cells as zero. Retaining the disclosure flag shows that suppression affects 68 percent of county-year records in data processing, 51 percent in telecommunications, and 66 percent in web and information services. Treating the suppressed cells as observed zeros would mechanically generate employment growth when a value first becomes publishable.

The second problem is that host counties were already changing before the facility opened. We estimate cohort-by-event-time effects relative to the year before opening, using not-yet-treated counties as controls and a state-cluster bootstrap [Callaway and Sant'Anna, 2021]. Table 5 reports the average estimate over event years zero through five together with a joint test of the pre-period leads. Four of the ten hyperscale outcome series change significantly before opening, with the failures concentrated in telecommunications, web and information services, and professional services. Data-processing establishments are the exception. Their pre-trend test does not reject, and the count rises when the facility enters the county.

Table 5: County-industry disclosure and pre-trend analysis for hyperscale facilities

Industry (NAICS)	Employment suppressed	Employment effect	Pre-trend p	Establishments effect	Pre-trend p
Data processing (518)	68%	+5.2%	0.58	+27.2%***	0.11
Telecommunications (517)	51%	+5.2%	0.28	−2.6%	0.04
Web and information services (519)	66%	+9.3%	0.39	+18.3%***	0.03
Computer systems design (5415)	40%	−0.9%	0.85	+5.7%	0.11
Professional and technical services (541)	27%	+6.2%**	< 0.01	+7.6%***	< 0.01

Notes: Employment suppression is the share of reported county-sector-year records carrying the BLS non-disclosure code. Effects are cohort-weighted mean log changes over event years zero through five, transformed to percent, relative to not-yet-treated counties. Pre-trend p is a joint test for event years −4 through −2. Inference uses 1,999 state-cluster bootstrap draws. The 519 series ends in 2021 because the 2022 NAICS revision moved major components to sectors 513 and 516, so its post-period average is taken over a shorter window than the other sectors and is weighted toward earlier cohorts. Stars refer to the post effect. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Benjamini–Hochberg q-values across the ten sector-outcome tests are < 0.001 for data-processing establishments, 0.011 for web and information-services establishments, 0.066 for professional-services employment, and < 0.001 for professional-services establishments.

Data-processing establishments rise by 27 percent, and the estimate survives a Benjamini–Hochberg adjustment across the ten sector-outcome tests. Because this sector contains the facility itself, the increase shows that the new data center appears in the county records. Once we retain the suppression flags, the employment estimates for data processing and telecommunications are too imprecise to distinguish from zero. Establishment counts in web and information services and in professional services also rise and survive the multiple-testing adjustment, but both series begin increasing before the facility opens. The county records therefore confirm the facility's entry but do not show that related industries grew in response.

6 Incidence outside the labor market

Within-site estimates place rent increases at +6.6% within 5 kilometers, +9.4% between 5 and 10 kilometers, and +8.5% between 10 and 25 kilometers. Unlike the capital footprint, this gradient does not decline with distance, raising the possibility that it reflects broader metropolitan rent growth rather than the facility. A density-matched comparison shows that similar locations without a data center also experienced substantial rent growth during the 2020–2023 work-from-home period. Facility-level synthetic controls yield smaller increases of two to five percent across the three rings, although the synthetic controls do not support formal placebo inference. The built-versus-near-miss estimate is +4.3% ($p = 0.23$), with a flat pre-period ($p = 0.44$). We therefore treat a rent increase of a few percent as suggestive and the larger within-site estimates as upper bounds. Colocation facilities show no detectable rent response.

Housing supply and net migration do not show a corresponding increase. In the Census Building Permits Survey, permits for buildings with at least five units rise by 3.4% ($p = 0.87$). For migration, we assign each county its first reliably dated hyperscale opening within 25 kilometers, including openings before the IRS panel, and compare treated counties with those not yet treated. Gross inflows rise by 5.4% ($p = 0.02$) and gross outflows by 7.6% ($p < 0.01$), but outflows and migrant income were already increasing before opening. We therefore do not interpret the gross-flow estimates as effects of the data center. The net measures are close to zero. The log inflow-to-outflow ratio moves by -1.4% over event years zero through five ($p = 0.52$), while the analogous adjusted-gross-income ratio moves by +0.4% ($p = 0.91$). Their pre-trend p-values are 0.14 and 0.09, respectively. Net migration therefore does not rise, and the pre-trends prevent a causal interpretation of the gross flows.

Foot traffic and commercial spending also show no robust response relative to near-miss sites. Foot traffic falls by roughly 12 percent in the full panel ($p = 0.12$), but the estimate attenuates to 3 percent in the balanced panel ($p = 0.68$). Commercial spending rises by 4.6% ($p = 0.60$). Although the within-site spending estimate is positive and passes a random-donor comparison, it is imprecise and is not confirmed by the built-versus-near-miss design.

The fiscal estimates are concentrated in school finance rather than broad local government spending. Table 6 uses the same not-yet-treated estimator and state-cluster bootstrap as the county analyses. Local school property-tax revenue rises by 16.6% ($p = 0.07$) without a detectable pre-trend, and capital outlay per pupil rises by 28.5%, although the latter estimate is imprecise ($p = 0.18$). Total local revenue and local revenue per pupil remain unchanged. County highway spending also has a positive, marginally significant estimate, but its magnitude changes materially when we update the opening dates, and the broader public-works series fails its pre-trend test. Neither the property-tax nor the highway estimate survives adjustment across the six fiscal outcomes ($q = 0.21$ for each). After this adjustment, neither fiscal estimate is statistically distinguishable from zero.

Table 6: Public-finance effects after hyperscale openings

Outcome	Geography	Effect	Post p	FDR q	Pre-trend p
Local property-tax revenue	school district	+16.6%	0.07	0.21	0.31
Total local revenue	school district	−2.3%	0.83	0.83	0.55
Local revenue per pupil	school district	+2.0%	0.76	0.83	0.14
Capital outlay per pupil	school district	+28.5%	0.18	0.35	0.57
Highway current + construction	county, balanced	+38.9%	0.07	0.21	0.15
Broad public works	county, balanced	+9.3%	0.40	0.61	< 0.01

Notes: Cohort-weighted mean log changes over event years zero through five, transformed to percent, relative to not-yet-treated geographies. School data are NCES/Census F-33 finance files for 2012–2023. County data are Census local government finance files for 2012–2023. Outcomes are adjusted for state-by-year means before the cohort comparisons. Inference uses 1,999 state-cluster bootstrap draws. Pre-trend p jointly tests event years -4 through -2. FDR q applies the Benjamini–Hochberg adjustment across the six outcomes in the table. The highway estimate moves materially across alternative opening dates and is treated as a sensitivity result rather than a headline effect.

The non-labor outcomes do not indicate a broad increase in local demand. Rents may rise modestly, while school revenue and road spending have positive but imprecise estimates. Net migration, new multifamily permits, foot traffic, and commercial spending do not rise.

7 Interpretation

Nighttime lights establish that construction occurred at the location and time recorded in our data and that the visible footprint ends within a few kilometers. The lights capture the facility itself, including security and parking lights, substations, and cooling equipment, rather than local economic activity. We use advertised salaries, hiring, and firm registrations to test whether the physical investment spreads beyond the site.

Advertised salaries and new-firm registrations do not rise at opening, and their upper confidence bounds are 5.5 and 1.6 percent. The firm-entry bound excludes a response of the size commonly associated with agglomeration, and the salary bound excludes the corresponding wage response unless little of any productivity gain reaches posted salaries (Section 5.4). The hiring confidence interval rules out increases above 17 percent, but supplier postings are too sparse to provide a useful bound. We can therefore rule out broad wage and firm-entry effects more confidently than gains among local suppliers.

Three limitations could allow a conventional local multiplier to go undetected. The response could occur beyond the zones we measure, although the 50 to 150 kilometer comparison is designed to capture that possibility. It could appear in employment that is neither advertised nor registered, some of which should nevertheless enter the county records. The response could also arrive outside our event windows, a possibility that future data on the post-2022 campuses will test. The geography of compute in Appendix A points instead to weak local linkages. Data centers and compute firms value the same metropolitan infrastructure and skilled labor, but 69 percent of compute firms within 25 kilometers predate the nearest facility. Their co-location therefore does not show that the data center created the surrounding cluster.

Synthetic-control rent estimates range from two to five percent, while net migration and multifamily permitting remain unchanged. School property-tax revenue and highway spending also have positive point estimates, although neither is statistically significant after correcting for the six fiscal outcomes we test. Residential electricity rates do not rise for the cloud-era cohorts in our sample. We do not observe land prices, environmental damages, or the terms of utility and tax agreements, so we cannot determine whether local welfare rises.

Finally, hyperscale and colocation facilities differ in both technology and location. Hyperscale campuses are generally larger and more likely to involve greenfield construction, whereas colocation facilities are multi-tenant and concentrated in denser markets. The hyperscale-colocation contrast describes the projects that jurisdictions encounter, but it combines facility and market differences. It does not identify the effect of randomly assigning one facility type rather than the other.

8 Conclusion

A hyperscale data center rivals the manufacturing plants in the agglomeration literature in capital scale, but the local response is much narrower. At isolated sites, nighttime lights rise when construction begins and fade within five kilometers, while daytime imagery confirms that the physical change starts with construction. Announced projects that were not built show no comparable break, although these near misses are not balanced runner-up sites.

The facility itself is visible in the data, but a surrounding cluster is not. Data-processing establishments rise when the facility opens. Advertised salaries and new-firm registrations do not rise, and the confidence intervals exclude increases above 5.5 and 1.6 percent. Compute firms are disproportionately likely to locate near data centers, but most arrived first, and exurban campuses often have few such firms nearby.

For the cloud-era cohorts we study, a large physical investment therefore does not produce the nearby worker and firm responses needed to support a broad local-multiplier claim. Rent and public-finance estimates are positive but imprecise. Two limits define the scope of this conclusion. The hiring and supplier-posting estimates allow for moderate gains, and the post-2022 artificial intelligence campuses are larger and more power-intensive than any facility in our sample. Whether those new campuses generate a different local response remains an open question, even as states continue to write subsidies for them.

A The geography of compute

A common rationale for data-center incentives is that a facility will anchor a local digital economy by attracting firms that use cloud capacity. We examine this claim by mapping the location of heavy compute users relative to data centers and asking whether any shared location predates the facility. We use the Veridion firmographic database [Veridion, 2026], which geolocates 6.5 million US establishments and classifies them by industry, employment, and founding year. We define the compute-native core to include software publishing, computing infrastructure and hosting, computer systems design, and web and streaming services. A broader definition adds finance, scientific R&D, and semiconductors.

A.1 Compute users cluster near data centers

Compute users are closer to data centers than the average firm. The median compute-native firm is 11 kilometers from the nearest data center, compared with 20 kilometers for non-compute firms. Similarly, 69 percent of compute users are within 25 kilometers of a data center, compared with 55 percent of other firms (Table A.1). Proximity also varies across compute-intensive industries. Cloud and hosting firms are nearest, at a median distance of 6 kilometers, followed by software firms at 9 kilometers. Telecommunications and streaming firms, whose operations are more geographically dispersed, are farther away.

The proximity of compute users to data centers is not an artifact of headquarters location. When we measure a firm's operating footprint using the geography of its job postings instead, compute users remain closer to data centers. The distances are somewhat larger because branch offices extend beyond the headquarters metropolitan area. The proximity gap also does not simply reflect the concentration of both groups in a handful of technology hubs. After removing the 45 largest metropolitan areas, compute users remain closer to the remaining data centers than non-compute firms are, with median distances of 75 and 87 kilometers, respectively. Software firms are among the closest, and the broader association remains visible in second-tier cities and rural areas.

Table A.1: Distance from firms to the nearest data center

	Firms	Median km	≤ 25 km	≤ 50 km
Panel A: Headquarters location (Veridion)				
Non-compute firms	5,692,641	20.5	55%	70%
Compute users (core)	309,760	11.0	69%	81%
Cloud and hosting	26,188	6.1	75%	84%
Software	101,757	8.9	72%	84%
Computer systems design	154,934	11.7	68%	81%
Finance	193,540	15.9	62%	75%
Telecommunications	10,125	22.2	53%	66%
Panel B: Establishment footprint (job postings)				
Compute users (core)		15.1	63%	76%
Panel C: Outside the 45 largest metros				
Non-compute firms	2,603,195	86.8	20%	33%
Compute users (core)	103,356	75.3	30%	41%

Notes: Great-circle distance from each firm to the nearest reliably dated hyperscale data center. Panel A places each firm at its Veridion headquarters coordinate. Panel B replaces the headquarters with the firm's job-posting locations (WageScape), weighted by the number of distinct hiring firms at each location, so it reflects the operating footprint rather than the headquarters. Panel C excludes all firms and data centers within 50 km of the 45 largest metropolitan areas and technology hubs and measures distance to the nearest remaining data center. Compute-native "core" sectors are software publishing, computing infrastructure and hosting, computer systems design, and web and streaming services.

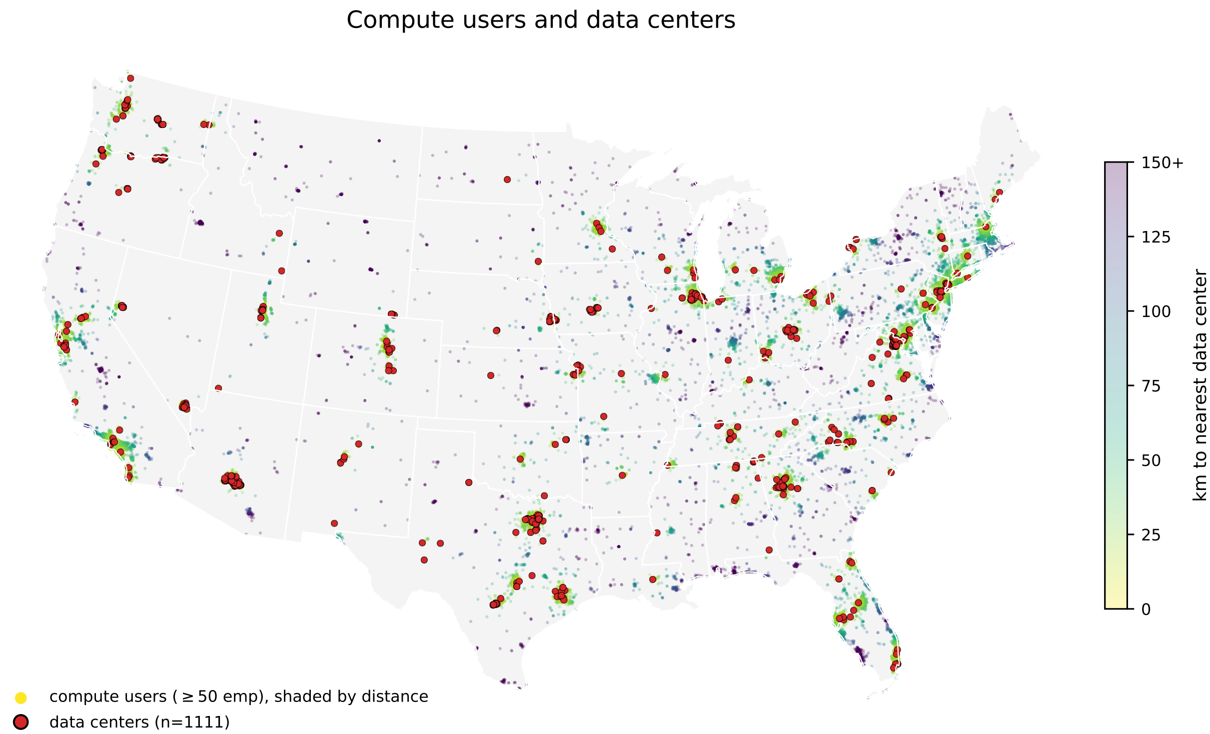

Figure A.1: Compute-user firms and data centers across the continental United States. Points are compute-native firms with at least 50 employees, shaded by distance to the nearest data center, with data centers marked in red. Both concentrate in the same coastal and metropolitan corridors. Points in the interior are darker because they are farther from a facility.

Alt text: A US map shows compute firms and data centers concentrated in coastal and metropolitan corridors, with sparse coverage across much of the interior.

The two groups trace the same map (Figure A.1), with both concentrated in coastal and metropolitan corridors and becoming less common across the interior. This association is strongest among large firms. As firm employment increases, the median distance to a data center falls monotonically, from 13 kilometers for the smallest firms to 4 kilometers for firms with more than ten thousand employees. The distribution narrows as well. Only about 5 percent of compute firms with at least a thousand employees are more than 50 kilometers from any data center. Those exceptions are generally software companies that grew in place in second-tier cities, such as Esri in Redlands, Jack Henry in rural Missouri, and Ansys outside Pittsburgh. Their experience nevertheless shows that even large compute users can access cloud capacity without locating near a data center.

A.2 Two kinds of data-center hub

The geographic overlap is primarily a metropolitan phenomenon. Grouping data centers into hubs reveals a sharp divide between metropolitan and exurban locations. Metropolitan hubs are embedded in dense clusters of sector-matched compute users. The 239 data centers around Ashburn, Virginia sit among thousands of government IT contractors, with DXC, Leidos, Booz Allen, SAIC, and CACI all within 25 kilometers. The Bay Area hubs are ringed by Google, Broadcom, Synopsys, and ServiceNow. By contrast, the large exurban hyperscale clusters are isolated. The 50 data centers around Hillsboro, Oregon and the cluster near Quincy, Washington each have on the order of four compute firms within 25 kilometers. The same type of physical infrastructure coincides with a dense compute neighborhood in a technology center and a sparse one in the exurbs.

A.3 Shared location does not mean that data centers created the cluster

Part of the explanation is shared infrastructure. Data centers and compute users both require access to the high-voltage grid, and both sit closer to it than ordinary firms. The median data center is 3.4 kilometers from a 230-kilovolt substation, the median compute user 5.0 kilometers, and the median other firm 6.4 kilometers. High-voltage substation density independently predicts where compute users locate. But the grid is only part of the story. Compute users remain closer to data centers after controlling for substation access. A shared pull toward the power backbone, fiber, and skilled labor can therefore place both in the same markets without one drawing the other.

Three patterns weigh against the interpretation that a data center creates the local compute cluster. First, the compute users near data centers predate them. Among compute firms within 25 kilometers of a data center, at least 69 percent were founded before the nearest data center opened. Because the firm registry over-represents recently founded firms, the share is likely higher. The local compute base usually existed first. Ashburn's history is consistent with this ordering. Northern Virginia's government technology firms and early internet exchange preceded the region's data-center cluster. Second, the job-posting analysis finds no detectable increase around opening, either in compute-using industries or overall, although the estimates are imprecise. Third, exurban data centers are located in areas with few compute firms. The median metropolitan data center has roughly 2,700 compute firms within 25 kilometers, while the median exurban data center has 41 (Table A.2).

Table A.2: Compute users near metropolitan versus exurban data centers

	Data centers	Median compute firms $\leq$ 25 km	Share with $<$ 5 nearby
Metropolitan data centers	824	2,702	0%
Exurban data centers	287	41	11%

Notes: Each data center is classified as metropolitan if it lies within 50 km of one of the 45 largest metropolitan or technology-hub centers, and exurban otherwise. The middle column reports the median count of compute-native firms (Veridion) within 25 km of the data center. The last column reports the share of data centers with fewer than five compute firms within 25 km. Separately, among compute firms within 25 km of any data center, 69 percent were founded before the nearest data center opened.

A.4 Implication for place-based policy

Data centers, especially exurban facilities, do not appear to draw compute users to the surrounding area. Firms that consume cloud capacity are attracted to transmission, fiber, and skilled labor, assets that are concentrated in metropolitan areas and are not created by a single facility. Most compute users located near data centers were already present, and we find no systematic increase in firm entry after opening. The capital footprint is large and local, while rent estimates are smaller and less precise. We do not detect the formation of a digital-firm cluster.

B The instrument search

A natural way to address endogenous site selection would be to find an instrument for data-center location. We investigated several candidates but did not find one with a credible exclusion restriction. This appendix documents that search and explains why the paper instead relies on the within-site and built-versus-near-miss designs. These tests also clarify the limits of common predictors of data-center siting.

Predetermined siting instruments. We tested several predetermined predictors of data-center siting, interacting them with the national capacity wave where appropriate. We also applied a manufacturing falsification test. A valid instrument for data-center activity should not independently predict manufacturing employment. Table B.1 summarizes the results.

Transmission-substation density strongly predicts siting, but it fails the falsification test because it also captures the availability of industrial land. The 1980 urban college share, which predicts entry in county-level studies of all data centers, has the opposite sign for hyperscale facilities. These campuses tend to locate on lower-college exurban land rather than in educated metropolitan centers. Proximity to long-haul fiber predicts siting across metropolitan areas but not within a county, so it cannot identify the sub-county effect studied here. Climate, measured by wet-bulb temperature and free-cooling days, does not predict siting. Operators can engineer around climate conditions when pursuing cheap power, and the largest US hyperscale hub is hot and humid. Finally, the timing of generation capacity has no predictive power within counties.

The same features that predict hyperscale siting, including power, industrial land, and fiber, are also correlated with a location's broader industrial character. By contrast, climate and historical human capital are less likely to violate the exclusion restriction but do not predict where hyperscale facilities locate. We therefore do not find a credible predetermined instrument for siting at the sub-county scale.

Table B.1: Predetermined siting instruments and why each fails

Instrument	First stage	Why it fails
345 kV substation density	strong	predicts manufacturing employment
1980 urban college share	moderate	predicts fewer hyperscale sites
InterTubes fiber proximity	metro-scale	no within-county effect
Climate (wet-bulb, free-cooling)	none	does not predict siting
Generation-capacity timing	none	does not predict siting within counties

Notes: Each instrument is the predetermined county feature interacted with the national hyperscale capacity wave. Predicting manufacturing employment violates the exclusion restriction.

A shift-share instrument also fails. We also construct a shift-share instrument similar to that used in concurrent county-level work [Alvarez et al., 2026], interacting predetermined fiber and college exposure with national growth in data-center demand. The instrument has predictive power. The first-stage F-statistic ranges from twenty to thirty when square footage is the endogenous variable. It does not, however, resolve the exclusion concerns described above. Public measures of data-center scale, including capacity, square footage, and facility count, are lumpy and supply-driven, while the underlying exposure measures also predict broader local development.

Because the shift-share design does not resolve the exclusion concerns, we do not use its estimates as evidence. The instrument also predicts population-serving sectors such as health, reinforcing the concern that it captures general area development. We therefore report the exercise as documentation of the search rather than as an identification strategy.

C Incidence on electricity rates

A central concern in data-center siting debates is that the load from a hyperscale facility will raise electricity bills for nearby households. Existing estimates are mixed. Wholesale-market modeling finds higher wholesale prices [Kay et al., 2026], utility-level estimates find higher retail rates, especially at investor-owned utilities [Meeks et al., 2026], and county-level instrumental-variable estimates find higher local electricity prices [Alvarez et al., 2026]. In contrast, Watten et al. [2026] estimate that data-center growth modestly reduced average retail rates from 2015 through 2024. We estimate whether residential customers in a host utility's territory pay more. Data centers select utilities with cheap power and available capacity, so a comparison of treated and untreated utilities must account for the facility's siting choice.

We treat each utility within each state as a separate unit. Using Energy Information Administration Form 861, we compute average residential prices as revenue over megawatt-hours sold from 2012 through 2024. A unit is treated in the year the first hyperscale facility opens in a county it serves. The hyperscale sample contains 52 utility-states first treated between 2015 and 2021, none of which also hosts a colocation facility. We compare them with roughly 2,000 utility-states that host no facility in the registry and, separately, with 54 utilities serving counties with an announced hyperscale near miss. A separate colocation sample of 60 treated utility-states is reported alongside.

Figure C.1 reports a cohort-by-event-time estimator using not-yet-treated controls. Against never-treated utilities, the mean effect over years zero through five is -1.0% ($p = 0.42$), with a joint pre-trend p-value of 0.97. The balanced panel, which retains utility-states observed in at least twelve of the thirteen years, gives -1.5% ($p = 0.17$). Against near-miss utilities, the full sample estimate is -0.9% ($p = 0.37$), and the year-five estimate is -1.9% ($p = 0.27$). Colocation estimates are small and positive, with a mean effect of $+0.7\%$ ($p = 0.54$).

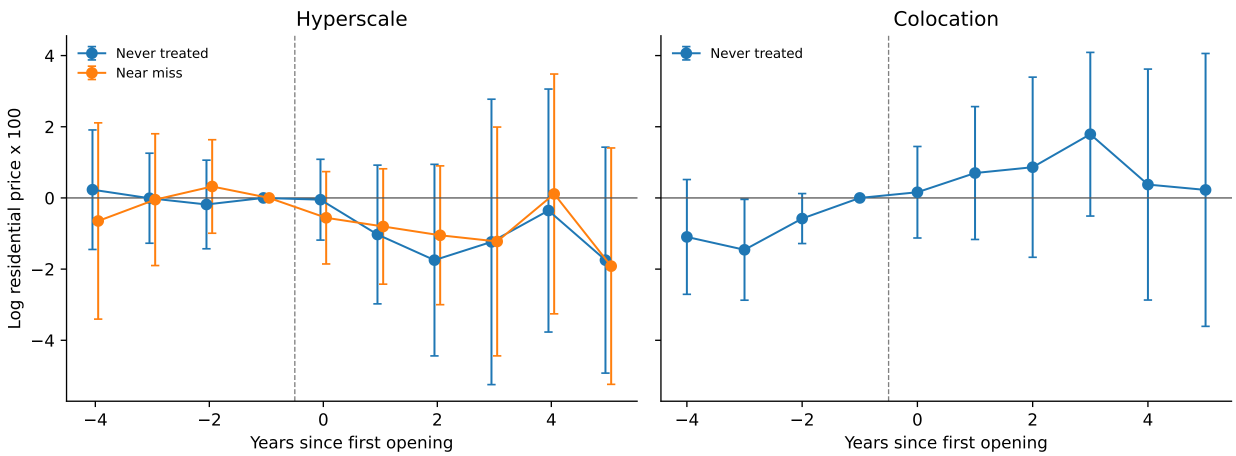

Figure C.1: Residential electricity prices around the first data-center opening in a utility's territory

Alt text: Two event-study panels show residential-price estimates around hyperscale and colocation openings. Hyperscale estimates become modestly negative after opening. Colocation estimates fluctuate around zero with wide intervals.

Notes: Cohort-by-event-time estimates of log average residential price (revenue over MWh, EIA Form 861 bundled-service utilities, 2012–2024). The hyperscale sample contains 52 treated utility-states, 2,040 never-treated controls, and 54 near-miss controls. The colocation sample contains 60 treated utility-states and the never-treated controls. The never-treated comparisons remove state-by-year means. The near-miss comparison removes year means. Inference uses 1,999 state-cluster bootstrap draws. The reference period is the year before opening, and 95% confidence intervals are shown.

The estimates provide no evidence that residential rates rise in the host utility over this horizon. One possible explanation for the negative point estimates is that a large new load spreads fixed network costs across more electricity sales. The full-sample and near-miss averages are imprecise, however, and we do not observe each utility's cost allocation or contract. The design therefore cannot identify that mechanism. Retail rates are also only one incidence margin. Electricity generation creates environmental costs that may be borne outside the host utility [Muller, 2026]. The sample covers cloud-era cohorts through 2021 followed to 2024, not the larger post-2022 artificial intelligence build-out.

D Supplementary results

Table D.1 reports confidence intervals and minimum detectable effects for the agglomeration estimates. Table 4 reports the capital footprint across estimators, transforms, and construction cohorts.

Table D.1: Precision of the agglomeration-channel estimates

Outcome	Comparison	Units	Estimate	95% CI	80% MDE
Advertised salary	0–25 vs 50–150 km	119	+1.3%	[-2.7, +5.5]	5.9%
Job postings	0–25 vs 50–150 km	120	+5.2%	[-5.3, +16.8]	16.2%
Supplier postings, 0–25 km	0–25 vs 50–150 km	120	+5.9%	[-15.0, +32.0]	37.0%
New firm registrations	Within-site trend break	117	-0.5%	[-2.6, +1.6]	3.0%
Business applications	National county	38	+1.4%	[-2.6, +5.7]	6.0%

Notes: Preferred estimate for each outcome, transformed from log points to percent. MDE is the minimum two-sided effect detectable with 80 percent power at the five-percent level, holding the reported standard error fixed. Salary, job postings, and supplier postings compare the pooled 0–25 km area with a 50–150 km comparison area, allowing labor-market effects to extend beyond the immediate site. Firm registrations use the Virginia and Texas within-site trend break. Business applications use the national county trend break. The unit count is facilities for the ring designs and treated counties for the business-application design.

The outcome panels do not cover identical facilities. Table D.2 therefore repeats the national facility-ring estimates on a common hyperscale-site universe. The salary and total-posting estimates barely change. Supplier postings remain positive but imprecise, foot traffic and spending remain statistically indistinguishable from zero, and rent remains positive but imprecise. The qualitative conclusions are therefore not driven by differences in facility coverage across these panels.

Table D.2: Common-site robustness for hyperscale facility-ring outcomes

Outcome	Outcome-specific coverage	Common-site coverage
Advertised salary	+1.3% [-2.7, +5.5] ($N = 119$)	+1.5% [-2.9, +6.1] ($N = 108$)
Job postings	+5.2% [-5.3, +16.8] ($N = 120$)	+5.8% [-5.8, +19.0] ($N = 109$)
Supplier postings	+5.9% [-15.0, +32.0] ($N = 120$)	+10.8% [-12.6, +40.5] ($N = 109$)
Foot traffic	+0.1% [-8.8, +10.0] ($N = 182$)	-5.6% [-14.9, +4.8] ($N = 106$)
Commercial spending	+5.9% [-10.6, +25.5] ($N = 113$)	-4.1% [-16.9, +10.8] ($N = 81$)
Residential rent	+7.4% [-0.9, +16.4] ($N = 180$)	+9.5% [-0.2, +20.1] ($N = 117$)

Notes: Each cell reports the estimated opening-year jump, its 95% confidence interval, and the number of facilities. The common-site column restricts every outcome to the same 125 hyperscale facilities that have high- or medium-confidence dates, opened by the first quarter of 2024, and appear in all five national facility-ring panels. Salary, job postings, and supplier postings use the 0–25 versus 50–150 km trend-break design. Foot traffic, spending, and rent use the 0–5 versus 25–50 km trend-break design with state-by-month fixed effects. These outcomes use the trend-break form because foot traffic and spending do not have the flat pre-period that the plain within-site comparison requires (Table 3). Standard errors are clustered by facility. Cell-specific counts can be lower than 125 when an outcome is missing in one of the required rings or periods. The capital, two-state registration, and county analyses retain their design-specific samples because they use different timing requirements or geographic units.

Data Availability Statement

This study combines publicly available data with several proprietary datasets obtained under license. All code that constructs the analysis datasets from the raw sources and reproduces every table and figure will be deposited in a public data and code repository. The deposit includes the publicly available inputs and all author-constructed files, namely the data-center registry, the operating and construction-start dates, and the announced near-miss cohort. The proprietary inputs cannot be redistributed. For each, we report the provider and access terms and include the exact product identifiers, date ranges, geographic filters, and extraction code, so that a licensed user can regenerate the facility-by-ring panels and reproduce all results.

Publicly available data.

- *Satellite nighttime lights.* VIIRS Day/Night Band Monthly Composite (NOAA, product VCMCFG), accessed through Google Earth Engine.
- *Daytime imagery for construction dating.* Sentinel-2 and Landsat surface reflectance (ESA and USGS), accessed through Google Earth Engine.
- *Data-center registry.* The starting point is the IM3 Open Source Data Center Atlas (Pacific Northwest National Laboratory), augmented by the authors from operator press releases, state economic-development announcements, the Good Jobs First subsidy tracker, trade press, DatacenterMap, Cloudscene, and data-center REIT filings. The compiled registry, dates, and announced near-miss cohort are in the deposit.
- *Construction timing.* Federal Aviation Administration Obstruction Evaluation / Airport Airspace Analysis filings (Form 7460-1).
- *Employment and establishments.* Quarterly Census of Employment and Wages (Bureau of Labor Statistics) and County Business Patterns (Census Bureau).
- *Migration and income.* IRS Statistics of Income county-to-county migration data.
- *Housing supply and population.* Census Building Permits Survey and 2020 Census Centers of Population (tract population-weighted centroids).
- *Workplace employment.* LEHD Origin-Destination Employment Statistics (LODES, Census Bureau).
- *Business formation.* Census Business Formation Statistics and state business-registration bulk files for Virginia, Texas, and Iowa (Secretaries of State and data.iowa.gov).
- *Local public finance.* Census of Governments and the Annual Survey of School System Finances (F-33).
- *Electricity.* U.S. Energy Information Administration Form 861.

- *Instrument-search inputs.* The InterTubes long-haul fiber map [Durairajan et al., 2015], the 1980 county urban-college share (IPUMS NHGIS), and transmission-substation locations (Homeland Infrastructure Foundation-Level Data).

Proprietary data (available under license, not redistributable).

- *Job postings.* LinkUp Job Records [LinkUp, 2025].
- *Advertised salaries.* WageScape salary postings [WageScape, 2022].
- *Residential rents.* RentHub Rental Data [RentHub, 2022].
- *Consumer spending.* SafeGraph Spend Patterns [SafeGraph, 2022].
- *Foot traffic.* Advan Weekly Patterns [Advan Research, 2025].
- *Firmographics.* Veridion firm master data.

The five Dewey-distributed datasets are available to researchers through a subscription to Dewey Data (`https://www.deweydata.io`). Veridion firmographics are available under license from Veridion (`https://veridion.com`). The authors accessed these data under institutional license and are not permitted to redistribute the raw files. The deposit provides the dataset identifiers and the extraction and aggregation code required to rebuild the analysis files from a licensed copy.

CRediT authorship contribution statement

Dany Bahar: Conceptualization, Methodology, Writing – review and editing. **Greg C. Wright:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review and editing.

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During the preparation of this work, the authors used OpenAI ChatGPT and Codex, Anthropic Claude, and Google Gemini to assist with code debugging, robustness checks, and editorial revision. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the article.

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